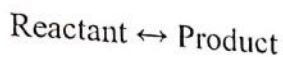


CHAPTER-7CHEMICAL EQUILIBRIUMREVERSIBLE REACTIONS**DEFINITION:**

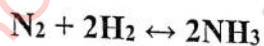
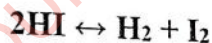
"Those chemical reactions which take place in both the directions and never proceed to completion are called Reversible reaction."

For this type of reaction both the forward and reverse reaction occurs at the same time so these reaction are generally represented as;



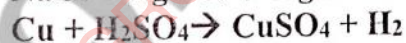
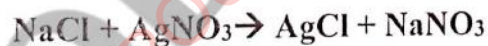
The double arrow $\leftrightarrow$ indicates that the reaction is reversible and that both the forward and reverse reaction can occur simultaneously.

Some examples of reversible reactions are given below.

EXAMPLEIRREVERSIBLE REACTIONS**DEFINITION:**

"Those reactions in which reactants are completely converted into product are called Irreversible reaction."

These reactions proceed only in one direction. Examples of such type of reaction are given below:

EXAMPLEEQUILIBRIUM STATE**DEFINITION:**

"The state at which the rate of forward reaction becomes equal to the rate of reverse reaction is called Equilibrium state."

EXPLANATION

Consider the following reaction;



It is a reversible reaction. In this reaction both the changes (i.e. forward & backward) occur

simultaneously. At initial stage reactant A & B are separated from each other therefore the concentration of C and D is zero.

When the reaction is started and the molecules of A and B react with each other the concentration of reactant is decreased while the concentration of product is increased. With the formation of product, the rate of forward reaction decreased with time but the rate of reverse reaction is increased with the formation of product C & D.

Ultimately a stage reaches when the number of reacting molecules in the forward reaction equalizes the number of reacting molecules in the reverse direction, so this state at which the rate of forward reaction becomes equal to the rate of reverse reaction is called equilibrium state.

LAW OF MASS ACTION

STATEMENT

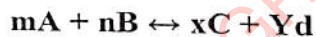
"The rate at which a substance reacts is proportional to its active mass and the rate of a chemical reaction is proportional to the product of the active masses of the reactant."

The term "active mass" means the concentration in terms of moles/dm³.

DERIVATION OF EQUILIBRIUM CONSTANT EXPRESSION

ASSUMPTION

Consider in a reversible reaction "m" mole of A and "n" moles of B reacts to give "x" moles of C and "y" moles of D as shown in equation.



EXPRESSION

In this process;

Or

The rate of forward reaction $\propto [A]^m [B]^n$

&

The rate of forward reaction = $K_f [A]^m [B]^n$

Or

The rate of reverse reaction $\propto [C]^x [D]^y$

But at equilibrium state

The rate of reverse reaction = $K_r [C]^x [D]^y$

Therefore,

Rate of forward reaction = Rate of reverse reaction

Or

$$K_f [A]^m [B]^n = K_r [C]^x [D]^y$$

Or

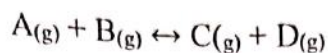
$$K_f / K_r = [C]^x [D]^y / [A]^m [B]^n$$

$$K_e = [C]^x [D]^y / [A]^m [B]^n$$

This is the expression for equilibrium constant which is denoted by K_c and defined as

"The ratio of multiplication of active masses of the products to the product of active masses of reactant is called equilibrium constant."

K_p In the case of gaseous equilibria, a partial pressure is used instead of concentration because at a given temperature partial pressure of a gas is directly proportional to concentration for this a constant is termed as K_p instead of K_c .



$$K_p = \frac{[P_c][P_d]}{[P_a][P_b]}$$

RELATIONSHIP B/W K_c , K_p , K_x , AND K_n

Despite the fact that the partial pressure of an ideal gas is directly proportional to its molar concentration at a constant temperature, the numerical value of K_c is often not equal to K_p . Accordingly, a quantitative relationship has been developed between these two equilibrium constants at a particular constant temperature.

$$K_p = K_c (RT)^{\Delta n}$$

Here,
 R = Gas constant which is taken in $0.082 \text{ atm dm}^3/\text{mol. K}$. Δn = [Sum of number of moles of products - the sum of the number of moles of reactants] in
 T = Absolute temperature the given balance chemical equation.

The relationship between K_p , K_x , and K_n may be written as

$$K_p = K_x (P)^{\Delta n}$$

$$K_p = K_n \left(\frac{P}{n}\right)^{\Delta n}$$

RELATION B/W K_c AND K_p :

- $K_p = K_c$; if there is no change in volume.
 $H_2 + I_2 \leftrightarrow 2HI$
- $K_p > K_c$; when reaction occurs with increase in volume.
 $2NH_3 \leftrightarrow N_2 + 3H_2$
- $K_p < K_c$; when the reaction occurs with a decrease in volume.
 $2SO_2 + O_2 \leftrightarrow 2SO_3$

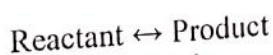
APPLICATIONS OF LAW OF MASS ACTION

There are two important applications of equilibrium constant.

1. It is used to predict the direction of the reaction.
2. K_c is also used to predict the extent of the reaction.

1. TO PREDICT THE DIRECTION OF REACTION:

The value of the equilibrium constant (K_c) is used to predict the direction of the reaction. For a reversible process;



With respect to the ratio of initial concentration of the reagent.

There are three possibilities for the value of K

1. It is greater than K_c
2. It is less than K_c
3. It is equal to K_c

Case I

If $[\text{Product}]_{\text{initial}} / [\text{Reactant}]_{\text{initial}} > K_c$, the reaction will shift towards the reverse direction.

Case II

If $[\text{Product}]_{\text{initial}} / [\text{Reactant}]_{\text{initial}} < K_c$, the reaction will shift towards the forward direction.

Case III

If $[\text{Product}]_{\text{initial}} / [\text{Reactant}]_{\text{initial}} = K_c$, This is equilibrium state for the reaction.

2. TO PREDICT THE EXTENT OF REACTION:

From the value of K_c we can predict the extent of the reaction.

CASE 1:

If the value of K_c is very large e.g.



From this large value of K_c it is predicted that the forward reaction is almost complete.

CASE 2:

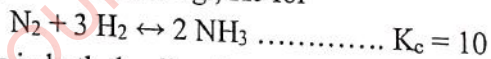
When the value of K_c is very low e.g.,



From this value it is predicted that the forward reaction proceeds with negligible speed i.e. backward reaction dominates.

CASE 3:

When the value of K_c is moderate, the reaction occurs in both the direction and equilibrium will be attained after certain period of time e.g., K_c for



so, the reaction occurs in both the direction.

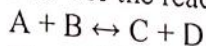
LE-CHATELIER'S PRINCIPLE**STATEMENT:**

"When a stress is applied to a system at equilibrium the equilibrium position changes so as to minimize the effect of applied stress."

The equilibrium state of a chemical reaction is altered by changing concentration pressure or temperature. The effect of these changes is explained by Le Chatelier.

EFFECT OF CONCENTRATION:

By changing the concentration of any substance present in the equilibrium mixture, the balance of chemical equilibrium is disturbed. For the reaction,



$$K_c = [\text{C}][\text{D}] / [\text{A}][\text{B}]$$

If the concentration of a reactant A or B is increased the equilibrium state shifts to right and yield of products increases.

But if the concentration of C or D is increased then the reaction will proceed in the backward direction with a greater rate and more A & B are formed.

EFFECT OF TEMPERATURE:

The effect of temperature is different for different type of reaction.
For an exothermic reaction the value of K_c decreased with the increase of temperature so the concentration of products decreases.

For an endothermic reaction heat is absorbed for the conversion of reactant into product so if temperature during the reaction is increased then the reaction will proceed with a greater rate in forward direction.

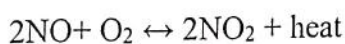
ENDOTHERMIC REACTION:

Temperature increase $\longrightarrow$ More products are formed
Temperature decrease $\longrightarrow$ More reactants are formed



EXOTHERMIC REACTION:

Temperature increase $\longrightarrow$ More reactants are formed
Temperature decrease $\longrightarrow$ More products are formed



EFFECT OF PRESSURE:

The state of equilibrium of gaseous reaction is distributed by the change of pressure. There are three types of reactions which show the effect of pressure change.

When the Number of Moles of Product are Greater:

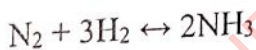
in a reaction such as



The increase of pressure shifts the equilibrium towards reactant side.

When the Number of Moles of Reactant are Greater:

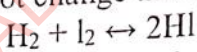
in a reaction such as



The increase of pressure shifts the equilibrium towards product side because the no. of moles of product are less than the no. of moles of reactant.

When Number of Moles of Reactants and Products are Equal:

These reactions where the number of moles of reactant are equal to the number of moles of product, a change of pressure does not change the equilibrium state e.g.,



Since the number of moles of reactants and products are equal in this reaction so the increase of pressure does not affect the yield of HI.

EFFECT OF CATALYST:

A catalyst minimizes the time to reach equilibrium but does not effect on change in equilibrium position of the system.

IMPORTANT INDUSTRIAL APPLICATION OF LE CHATELIER'S PRINCIPLE:

HABER'S PROCESS:

This process is used for the production of NH_3 by the reaction of nitrogen and hydrogen. In this process 1 volume of nitrogen is mixed with three volumes of hydrogen at 500°C and 200 to 1000 atm pressure in presence of a catalyst



1. Effect of Concentration

The value of K_c for this reaction is

$$K_c = [\text{NH}_3]^2 / [\text{N}_2] [\text{H}_2]^3$$

Increase in concentration of reactants which are nitrogen and hydrogen the equilibrium of the process shifts towards the right so as to keep the value of K_c constant. Hence the formation of NH_3 increases with the increase of the concentration of N_2 or hydrogen.

2. Effect of Temperature

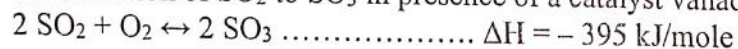
It is an exothermic process, so heat is liberated with the formation of product. Therefore, according to Le Chatelier's principle at low temperature the equilibrium shifts towards right to balance the equilibrium state so low temperature favour the formation of NH_3 .

3. Effect of Pressure

The formation of NH_3 proceeds with the decrease in volume, therefore, the reaction is carried out under high pressure or in other words high pressure is favorable for the production of NH_3 .

CONTACT PROCESS:

The process is used to manufacture H_2SO_4 on large scale. In this process the most important step is the oxidation of SO_2 to SO_3 in presence of a catalyst vanadium pentaoxide.



1. Effect of Concentration

The value of K_c for this reaction is

$$K_c = [\text{SO}_3]^2 / [\text{SO}_2]^2 [\text{O}_2]$$

Increase in concentration of SO_2 or O_2 shifts the equilibrium towards the right and more SO_3 is formed.

2. Effect of Temperature

Since the process is exothermic, so low temperature will favor the formation of SO_3 . The optimum temperature for this reaction is 400 to 450°C .

3. Effect of Pressure

In this reaction decrease in volume takes place so high pressure is favorable for the formation of SO_3 .

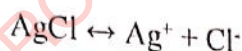
SOLUBILITY PRODUCT:**DEFINITION**

"The product of the concentration of ions in the saturated solution of a sparingly soluble salt is called solubility product."

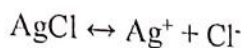
It is denoted by " K_{sp} ".

DERIVATION

When a slightly soluble ionic solid such as silver chloride is dissolved in water, it decompose into its ions



These Ag^+ and Cl^- ions from solid phase pass into solution till the solution becomes saturated. Now there exists an equilibrium between the ions present in the saturated solution and the ions present in the solid phase, thus



Applying the law of mass action

$$K_c = [\text{Ag}^+][\text{Cl}^-] / [\text{AgCl}]$$

Since the concentration of solid AgCl in the solid phase is fixed, no matter how much solid is present in contact with solution, so we can write.

$$K_c = [\text{Ag}^+][\text{Cl}^-] / K$$

Or

$$K_c \times K = [\text{Ag}^+][\text{Cl}^-]$$

Or

$$K_{sp} = [\text{Ag}^+][\text{Cl}^-]$$

The value of the solubility product is constant for a given temperature.

APPLICATION OF K_{sp}

Precipitation is a kind of double displacement reaction in which two solutions of different salts mixed together to form two products, one of these product is insoluble in solution and called as precipitate. For example when an aqueous solution of silver nitrate and potassium chloride mix together, the Ag^+ ions of AgNO_3 combine with Cl^- ions of KCl and form an insoluble silver chloride (AgCl) which can be precipitated out.



The net ionic equation of the above precipitation reaction is shown as,



The K_{sp} of AgCl tells us about its limit of solubility in solution but the question is whether or not a precipitate of AgCl forms when aqueous solutions of AgNO_3 , and KCl of specific concentrations mixed together. To reach the answer we should know another term known as ionic product.

Ionic product or reaction quotient is "the product of molar concentration of the ions each raised to the power by its coefficient mentioned in net ionic equation". Ionic product is relevant to both saturated and unsaturated solution while k_{sp} is only applicable to saturated solution which involves a dynamic equilibrium between an insoluble salt and its aqueous ions.

Now by using the following three relationships between ionic product (Q_{sp}) and solubility product (K_{sp}), one can predict even if the salt form precipitate by mixing two different salt solution.

$K_{sp} > Q_{sp}$ Solution is unsaturated in which no ppt is formed

$K_{sp} = Q_{sp}$ Solution is saturated

$K_{sp} < Q_{sp}$ Solution is supersaturated in which ppt is formed

COMMON ION EFFECT

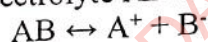
STATEMENT

"The decrease in the solubility of the salt in a solution that already contains an ion common to that salt, is known as common ion effect."

EXPLANATION:

In the solution of an electrolyte in water, there exist equilibrium between the ions and the dissociated molecules to which the law of mass action can be applied.

Considering the dissociation of an electrolyte AB we have

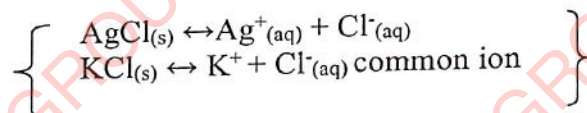


And

$$\frac{[A^+][B^-]}{[AB]} = K_{\text{(dissociation constant)}}$$

If now another electrolyte yielding A^+ or B^- ions be added to the above solution, it will result in the increase of concentration of the ions A^+ or B^- and in order that K may remain the same, the concentration AB must evidently increase. In other words the degree of dissociation of an electrolyte is suppressed by the addition of another electrolyte containing a common ion. This phenomenon is known as common ion effect.

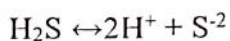
For E.g:



APPLICATION OF COMMON ION EFFECT

1. PRECIPITATION OF SULPHIDES GROUP OF II:

Sulphides of this group are precipitated by passing H_2S gas through the acidified original solution.



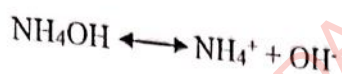
HCl provides H^+ ion as a common ion which shifts the equilibrium towards left. The concentration of common ion in solution increase and the concentration of S^{2-} ions is just enough to exceed the solubility product of II group sulphides. In this way the cation of II group are precipitated as CuS , PbS , CdS .

2. PRECIPITATION OF SULPHIDES OF GROUP IV:

The sulphides of this group are precipitated by H_2S gas in presence of NH_4OH . The OH^- ion of NH_4OH combines with H^+ of H_2S to form H_2O ($H^+ + OH^- \longrightarrow H_2O$). the removal of H^+ ion from the product side results in the increase concentration of sulphide ion S^{2-} which is enough to exceed the solubility product for the precipitation of CuS , NiS and ZnS etc.

3. PRECIPITATION OF SULPHIDES OF GROUP IV:

The cations of group III are precipitated by NH_4OH in presence of NH_4Cl . The common NH_4^+ ion suppresses the ionization of NH_4OH .



The common NH_4^+ ion shift the equilibrium towards left and concentration of OH^- decreases. Under these conditions, the solubility product of the hydroxides of Al, Fe and Cr is only exceeded due to which they are precipitated while the hydroxides of Zn, Ni and Co etc. are not precipitated due to their high solubility.

MULTIPLE CHOICE QUESTIONS

1. At equilibrium the rate of forward reaction and the rate of reverse reaction are _____.

(Equal, Changing, Different)

2. Such reactions, which proceed to forward direction only and are completed after sometime are called _____ reaction.

(Irreversible, Reversible, Molecular)

3. Such reactions, which proceed to both the direction and are never completed, are called _____ reaction.

(Irreversible, Reversible, Molecular)

4. The rate of chemical reaction is directly proportional to the product of the molar concentration of _____.

(Reactants, Products, Both reactants and products)

5. "If a system in equilibrium is subjected to a stress, the equilibrium shifts in a direction to minimize or undo the effect of this stress. This principle is known as _____.

(Le-Chatelier's Principle, Gay Lussac's Principle, Avogadro's Principle)

6. A very large value of K_c indicates that reactants are _____.

(Very stable, unstable, moderately stable)

7. A very low value of K_c indicates that reactants are _____.

(Very stable, very unstable, moderately stable)

8. The equilibrium in which reactants and products are in single phase is called _____.

(Homogenous Equilibrium, Heterogenous Equilibrium, Dynamic Equilibrium)

9. The equilibrium in which reactants and products are in more than one phases are called _____.

(Homogeneous Equilibrium, Heterogenous Equilibrium, Dynamic Equilibrium)

10. Chemical Equilibrium is _____ equilibrium.

(Dynamic, Static, Heterogeneous)

11. In exothermic reaction, lowering of temperature will shift the equilibrium to _____.

(Right, left, equally on both the direction)

12. In endothermic reaction, lowering of temperature will shift the equilibrium to _____.

(Right, left, equally on both the direction)

13. A catalyst _____ the energy of activation.

(Increases, decreases, has no effect on)

14. At equilibrium point _____.

(Forward reaction is increased, backward reaction is increased, forward and backward reactions become equal)

15. NH_3 is prepared by the reaction $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ $\Delta H = -21.9 \text{ Kcal}$. The maximum yield of NH_3 is obtained _____.

(At low temperature and high pressure, at high temperature and low pressure, at high temperature and high pressure)

16. When a high pressure is applied to the following reversible process: $\text{N}_2 + \text{O}_2 \rightleftharpoons 2\text{NO}$ The equilibrium will _____.

(shift to the forward direction, shift to the backward direction, not change)

17. The value of K_c _____ upon the initial concentration of the reaction.

(depends, partially depends, does not depend)

18. While writing the K_c expression, the concentration of _____ are taken in the numerator.

19. Solubility product constant is denoted by _____.

(K_c , K_{sp} , K_r)

20. "The degree of ionization of an electrolyte is suppressed by the addition of another electrolyte containing a common ion." This phenomenon is called _____.

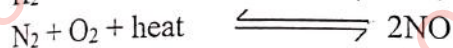
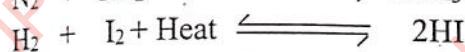
(Solubility Product, Common Ion Effect, Le-Chatelier's Principle)

THEORY QUESTIONS

- State and explain the law of MASS ACTION. Derive the expression for K_c of the following reaction:



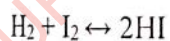
- Explain Reversible reaction and Equilibrium constant.
- Write relationship between K_c and K_p .
- How is K_c used to predict the direction of a reversible reaction?
- What is the chemical equilibrium? Illustrate it by the help of graph.
- How is the equilibrium constant value used to predict the extent of reaction?
- State and explain Le-Chatelier's principle.
- Discuss the condition to increase the yield of NH_3 in Haber's process.
- Predict the effect of change of temperature and effect of pressure on following equilibrium mixture:



- Define Solubility and Solubility product. Write the solubility product along with their units in MgSO_4 and $\text{Mg}(\text{OH})_2$.
- What is Common ion Effect? How is it applied in qualitative analysis?

NUMERICALS FROM PAST PAPERS**NUMERICALS OF K_c :**

- 3 moles of H_2 and 4 moles of I_2 were heated in a sealed tube at a given temperature at which K_c is 50. If the volume of tube is 1 dm^3 , determine the composition of the equilibrium.



$$x = 2.61 \text{ M}$$

- K_c for the reaction $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \leftrightarrow \text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O}$ at room temperature is 4. Calculate the equilibrium concentration of $\text{CH}_3\text{COOC}_2\text{H}_5$ when 1.66 moles of CH_3COOH and 2.17 moles of $\text{C}_2\text{H}_5\text{OH}$ are allowed to come to equilibrium.

$$x = 1.24 \text{ M}$$

- When the equilibrium was attained for the reaction: $\text{A} + \text{B} \leftrightarrow 2\text{C}$, the concentration of $[\text{A}] = [\text{B}]$ was 4 moles/dm^3 and that of C was 6 moles/dm^3 . Calculate K_c and initial concentration of A and B .

$$K_c = 2.25, \text{ A} = \text{B} = 7 \text{ mol/dm}^3$$

- In a reaction $\text{A} + \text{B} \leftrightarrow 2\text{C}$, 7 moles/dm^3 of A and 7 moles/dm^3 of B were mixed and allowed to attain equilibrium. If $K_c = 2.25$, find out the concentration of A , B and C at equilibrium.

- The equilibrium constant for the reaction $\text{N}_2 + \text{O}_2 \leftrightarrow 2\text{NO}$ at 2000°C is 0.1. Calculate the equilibrium concentration of reactants and products when the initial concentration of N_2 and O_2 are 10 moles/dm^3 each.

$$x = 1.36 \text{ M}$$

6. For the reaction $2\text{HI} \leftrightarrow \text{H}_2 + \text{I}_2$, one mole of HI was introduced into a vessel at constant temperature. Calculate the number of moles of I_2 when the equilibrium is reached. ($K_c = 0.0156$).
 $x = 0.099 \text{ M}$
7. In the reaction $\text{A} + \text{B} \leftrightarrow 2\text{C}$, when equilibrium is attained, the concentration of $\text{B} = 4 \text{ moles/dm}^3$ and $\text{C} = 6 \text{ moles/dm}^3$. Calculate the initial and equilibrium concentration of A. $K_c = 2.25$.
 $A_{\text{initial}} = 7 \text{ M}, A_{\text{eq}} = 4 \text{ M}$
8. 60 gm of Acetic acid and 46 gm of ethyl alcohol are mixed with each other at constant temperature and are allowed to attain the equilibrium. At equilibrium 58.2 gm of ethyl acetate and 12 gm of water were present. Find the equilibrium constant.
 $K_c = 0.4356$
9. 0.2 moles of A and 0.4 moles of B were reacted at certain temperature and were allowed to come to equilibrium: $\text{A} + \text{B} \leftrightarrow 2\text{B}$, the equilibrium mixture contained 0.1 moles A, find K_c if the volume of the container is 2.0 dm^3 .
 $K_c = 1.33$
10. A quantity of PCl_5 was heated in a 6 dm^3 flask at 250°C , till the following equilibria was established:
 $\text{PCl}_5 \leftrightarrow \text{PCl}_3 + \text{Cl}_2$, the equilibrium mixture contains 0.16 moles of PCl_5 , 0.16 moles of Cl_2 and 0.105 moles of PCl_3 , calculate for the equilibrium constant.
11. For the reaction $\text{PCl}_5 \leftrightarrow \text{PCl}_3 + \text{Cl}_2$, the equilibrium mixture at a definite temperature has the partial pressure of gases as $P_{\text{PCl}_5} = 0.36 \text{ atm}$, $P_{\text{PCl}_3} = 0.82 \text{ atm}$, $P_{\text{Cl}_2} = 0.82 \text{ atm}$, calculate K_p for the reaction.
 $K_p = 1.867$
12. One mole of PCl_5 was introduced in a vessel of 10 dm^3 capacity at constant temperature. At equilibrium 0.465 mole of Cl_2 gas were present, calculate the K_c for the reaction.
 $\text{PCl}_5 \leftrightarrow \text{PCl}_3 + \text{Cl}_2$
 $K_c = 0.404$
13. 1.5 moles of acetic acid and 1.5 moles of ethyl alcohol are mixed with each other at constant temperature and are allowed to attain the equilibrium. At equilibrium one mole of ethyl acetate was present, in one liter of the equilibrium mixture, find the equilibrium constant.
 $K_c = 4$
14. For the reaction $\text{H}_2 + \text{I}_2 \leftrightarrow 2\text{HI}$, $K_c = 49$. Calculate the concentration of HI at equilibrium when initially one mole of H_2 is mixed with one mole of I_2 in one liter flask.
 $[\text{HI}] = 0.777$
15. At equilibrium 12 liter flask contains 0.21 moles of PCl_5 , 0.32 moles of PCl_3 , 0.32 moles of Cl_2 . find the value of K_c for the reaction. $\text{PCl}_5 \leftrightarrow \text{PCl}_3 + \text{Cl}_2$
 $K_c = 0.04061$
16. 5.88 moles of Nitrogen and 16.2 mole of oxygen are mixed and heated at 2000°C until the equilibrium is established, 11.28 moles of Nitric oxide are formed. Calculate the value of equilibrium constant.
 $\text{N}_2 + \text{O}_2 \leftrightarrow 2\text{NO}$
 $K_c = 42.33$

APPLICATION OF K_c :

1. The K_c for the reaction $2\text{HI} \leftrightarrow \text{H}_2 + \text{I}_2$ is 1.3×10^{-2} . If there are $0.5 \text{ moles/dm}^3 \text{ H}_2$, $1.5 \text{ moles/dm}^3 \text{ I}_2$ and $5 \text{ moles/dm}^3 \text{ HI}$, predict the direction in which the reaction moves so as to achieve the equilibrium.
backward direction
2. K_c for the reaction is 0.0194 and the calculated ratio of the concentration of the reactant and product is 0.0116. Predict the direction of reaction.

SOLUBILITY PRODUCT:

1. The solubility of AgCl at 25 °C is $1.4 \times 10^{-3} \text{ g/dm}^3$. Its molecular mass is 143.5. calculate the :
 - a. Molarity of AgCl solution
 $9.75 \times 10^{-6} \text{ mol/dm}^3, 9.5 \times 10^{-11} \text{ M}^2$
 - b. solubility product of AgCl
2. The solubility of calcium oxalate CaC_2O_4 is 0.0116 gm/dm^3 at 25 °C. Find the solubility product of calcium oxalate.
 $\text{CaC}_2\text{O}_4 \leftrightarrow \text{Ca}^{+2} + \text{C}_2\text{O}_4^{-2}$
 Write the solubility product expression (K_{sp}) expression of the following.
 - a. CaCO_3
 $K_{sp} = 8.212 \times 10^{-9} \text{ M}^2$
 - b. Mg(OH)_2
3. The given reaction is, $\text{PbI}_2 \leftrightarrow \text{Pb}^{+2} + 2\text{I}^-$
 At 25°C the solubility of $\text{PbI}_2 = 0.63 \times 10^{-3} \text{ mol/dm}^3$; find the value of K_{sp} and express its units.
 $K_{sp} = 1.00 \times 10^{-9} \text{ mol}^3/\text{dm}^9$
4. The solubility of AgCl at 25°C is $1.4 \times 10^{-3} \text{ g/dm}^3$. What is its solubility product K_{sp} ? Atomic mass of Ag=108, Cl=35.5
 $K_{sp} = 9.5 \times 10^{-11} \text{ M}^2$
5. Find the K_{sp} of CaCO_3 , the solubility of CaCO_3 is 0.001 g/dm^3 .
 $\text{CaCO}_3 \leftrightarrow \text{Ca}^{+2} + \text{CO}_3^{-2}$
6. Will PbCrO_4 precipitate from a solution prepared by mixing 200 cm^3 of $2.5 \times 10^{-4} \text{ M Pb(NO}_3)_2$ and 600 cm^3 of $1.5 \times 10^{-8} \text{ M K}_2\text{CrO}_4$? (K_{sp} of $\text{PbCrO}_4 = 1.8 \times 10^{-14}$)
 $K_{sp} = 7.03 \times 10^{-13}$, ppt will form
7. The solubility product of AgCl is $9.6 \times 10^{-11} \text{ mol}^2/\text{dm}^6$ at 25°C. Calculate the solubility of AgCl in gm/dm^3 .
 (Ag=108 amu, Cl=35.5 amu)
8. What is the solubility of lead iodide in mol/dm^3 at 25°C? K_{sp} for PbI_2 is $1 \times 10^{-9} \text{ mol}^3/\text{dm}^9$.
9. The solubility product of BaSO_4 is $1.0 \times 10^{-10} \text{ mol}^2/\text{dm}^6$. Work out the solubility of the salt in gm/dm^3 .
 $x = 2.33 \times 10^{-3} \text{ gm/dm}^3$
10. Should PbCrO_4 precipitate from a solution prepared by mixing 100 cm^3 of $2.0 \times 10^{-6} \text{ M Pb(CH}_3\text{COO)}_2$ and 900 cm^3 of $1.5 \times 10^{-8} \text{ M Na}_2\text{CrO}_4$? (K_{sp} of $\text{PbCrO}_4 = 1.8 \times 10^{-14}$)
 $K_{sp} = 2.7 \times 10^{-15} \text{ M}^2$, ppt will not form
11. Should AgCl precipitate from a solution prepared by mixing 400 cm^3 of 0.1 M NaCl and 600 cm^3 of 0.03 M AgNO_3 solution? (K_{sp} of AgCl = $1.6 \times 10^{-10} \text{ mol}^2/\text{dm}^6$)
 $K_{sp} = 7.2 \times 10^{-4} \text{ M}^2$, ppt will form
12. The solubility of AgCl at 25°C is $1.4 \times 10^{-3} \text{ g/dm}^3$. Its molecular mass is 143.5 amu. Find
 - a. Molarity of AgCl
 - b. solubility product of AgCl
13. Calculate the solubility product of PbCrO_4 when solubility of PbCrO_4 is $1.0 \times 10^{-3} \text{ gm/dm}^3$.